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Blood flow throughout the circulatory system is dictated by blood (hydrostatic) pressure, vascular resistance (force opposing blood flow through a vessel), and cardiac output (blood volume expelled from the ventricles per unit time). When blood traverses a vessel, it exerts hydrostatic pressure on the vessel walls, which results in the forced movement of fluid out of vessels and into the interstitial space. The circulating plasma proteins cause the osmotic pressure within the vessel to be higher than that of the interstitial fluid. In turn, osmotic pressure causes fluid to flow from the interstitial space into blood vessels, opposing hydrostatic pressure.

Cardiac output depends in part on heart rate, which is tightly regulated by the sinoatrial (SA) and atrioventricular (AV) nodes, specialized groups of self-depolarizing cells located in the upper right atrium wall and lower interatrial septum, respectively. Action potentials (APs) generated by SA nodal cells stimulate atrial contraction as they travel to the AV node. The AV node delays AP transmission to ventricular cells, ensuring ventricular filling is complete prior to heart contraction.

A 63-year-old man was admitted to the hospital after he collapsed while exercising. The patient had an elevated heart rate and an abnormally low blood oxygen level on admission. X-rays revealed excess fluid in his lungs.

The systemic blood pressure gradient (ΔP) is equal to the mean pressure of blood exiting the heart through arteries minus the mean pressure of blood returning to the heart through veins. The relationship of ΔP to cardiac output (CO) and vascular resistance (VR) is given by:

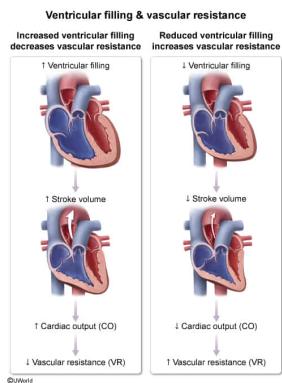
$$\Delta P = CO \times VR$$

If a change in CO caused a dramatic increase in a patient's VR, the patient's CO most likely had:

- ☐ A. increased due to higher volumes of blood pumped per minute. [20%]
- ☐ B. increased due to a higher rate of heart contraction. [11%]
- ☒ C. decreased due to reduced ventricular filling prior to contraction. [58%]
- ☐ D. decreased due to increased stroke volume. [9%]

Correct
58% answered correctly

Explanation:



The passage states that blood flow through the circulatory system is dictated by **blood** (hydrostatic) **pressure**, **vascular resistance (VR)**, force opposing blood flow through a vessel), and **cardiac output (CO)**, blood volume expelled from the ventricles per unit time). Generally, blood flows from high to low pressure; therefore, a pressure gradient also affects blood flow.

The question states that the systemic **blood pressure gradient (ΔP)** is calculated by subtracting the mean pressure of blood returning to the heart (venous pressure) from the mean pressure of blood exiting the heart (arterial pressure). Accordingly, the relationship of ΔP to CO and VR is given by:

$$\Delta P = CO \times VR$$

Manipulation of this equation gives the relationship of VR to ΔP and CO:

$$VR = \frac{\Delta P}{CO}$$

Consequently, an increase in this patient's VR resulted from an **increase in ΔP** and/or a **decrease in CO**. Specifically, CO is dependent on both **heart rate** (number of times a person's heart beats per minute) and **stroke volume (SV)**, blood volume expelled per left ventricle contraction). Stroke volume depends on the amount of blood loaded into the left ventricle prior to contraction. **Reduced ventricular filling** leads to less blood being expelled from the ventricle with each contraction (ie, reduced SV) and **decreased CO**. Therefore, a dramatic increase in this patient's VR most likely was caused by **decreased CO** due to **reduced ventricular filling prior to contraction**.

(Choices A and B) Higher volumes of blood pumped per minute (ie, increased SV) or a higher rate of heart contraction increases CO. However, an increase in this patient's CO would likely have **decreased, not increased, VR**.

(Choice D) Increased SV would have caused an increase, *not* a decrease, in CO.

Educational objective:

Blood flow through the circulatory system is affected by blood pressure, vascular resistance (VR, force opposing blood flow through a vessel), and cardiac output (CO, blood volume expelled from the ventricles per unit time). The relationship of these factors in the systemic circulatory system is given by: $\Delta P = CO \times VR$.

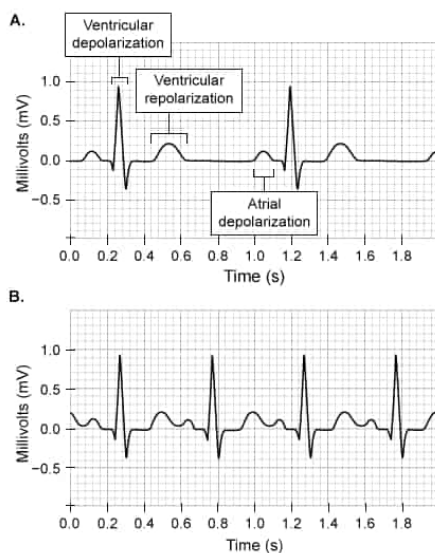
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The attending physician orders an ECG to measure the electrical activity of the patient's heart. The ECG of a healthy individual at rest (A) and the ECG of the patient at rest (B) are shown below.



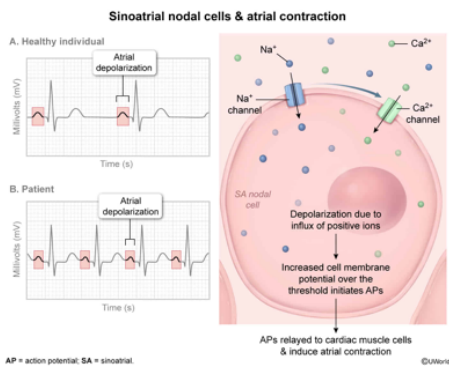
Which of the following conclusions regarding the patient's cardiac function is best supported by these results?

- ☐ A. Atrial contraction occurs less frequently in the patient than in the healthy individual. [4%]
- ☐ B. APs are fired less frequently from AV nodal cells in the patient than in the healthy individual. [4%]
- ☐ C. Ventricular depolarization occurs for a longer time period in the patient than in the healthy individual. [4%]
- ☒ D. Influx of positive ions into SA nodal cells occurs more rapidly in the patient than in the healthy individual. [86%]

Correct

86% answered correctly

Explanation:



As the passage states, heart rate is tightly regulated by the SA and AV nodes, specialized clusters of **self-depolarizing** cells found in the upper atrial wall and lower interatrial septum, respectively. Normally, **depolarization** occurs via the **influx of positive ions** (ie, Na^+ , Ca^{2+}) **into SA and AV nodal cells**, *increasing* cell membrane potentials and generating APs. APs fired from the SA node travel through cardiac muscle cells of the atria, inducing atrial contraction and ventricular filling. At the AV node, relay of APs to ventricular cells is delayed, ensuring that ventricular filling is complete prior to ventricular contraction. Next, APs travel through both ventricles, causing them to contract and expel blood into the arteries.

The electrical activity of the heart is often measured using an **ECG** machine, a device that records APs generated by SA and AV nodal cells as well as cardiac muscle cells. In this scenario, the physician orders an ECG to measure the electrical activity of the patient's heart. Compared to a healthy individual's ECG, the patient's ECG shows *more* waves of atrial and ventricular depolarization over the same time period. This indicates an *increased* number of APs fired from the patient's SA and AV nodal cells, leading to *more frequent* atrial and ventricular contraction (**Choices A and B**). Because depolarization and AP generation depend on the influx of positive ions into cells, a **more rapid influx of positively charged ions** into SA nodal cells may lead to *increased* firing of APs from the SA node and more *frequent* atrial contraction, as observed in the patient's ECG.

(Choice C) Although ventricular depolarization and contraction are occurring *more frequently* in the patient, the ECGs show that the **duration** of each ventricular depolarization event is the *same* in both the patient and the healthy individual.

Educational objective:

Heart rate is regulated by the activity of specialized clusters of self-depolarizing cells known as the SA and AV nodes. APs are initiated in the SA node and travel through the atria, stimulating atrial contraction. These APs then reach the AV node and, after a brief delay, are relayed to ventricular cells, stimulating ventricular contraction.

Time spent: 0 sec
Subject:
 Biology

QID: 402689
System:
 Circulation and Respiration

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As shown below, CO_2 reacts with water in the blood to form H_2CO_3 , a weak acid that can reversibly dissociate to form HCO_3^- and a free H^+ ion.



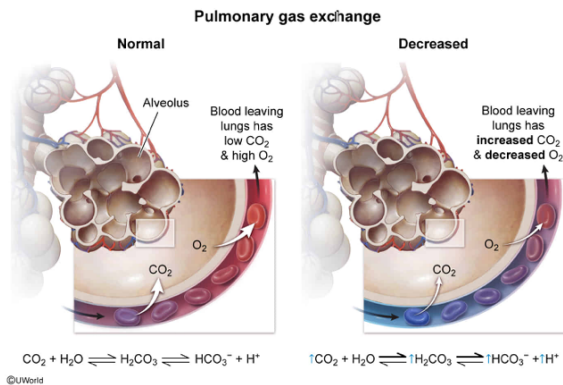
The patient is noted to have decreased pulmonary gas exchange. Which of the following will most likely occur in the patient's blood if this condition is left untreated?

- ✔ ☒ A. More H_2CO_3 will be produced and blood pH will decrease. [66%]
- ☐ B. Less H_2CO_3 will be produced and blood pH will increase. [16%]
- ☐ C. More H^+ ions will be present and blood pH will increase. [11%]
- ☐ D. Fewer H^+ ions will be present and blood pH will decrease. [6%]

Correct

66% answered correctly

Explanation:



Aerobic respiration in cells uses oxygen (O₂) obtained from the bloodstream to produce energy (ATP) from glucose, releasing carbon dioxide (CO₂) as a by-product. As blood flows through systemic capillaries, O₂ **diffuses** into tissues and CO₂ diffuses from tissues into blood. Veins transport deoxygenated blood containing CO₂ back to the heart. From the heart, deoxygenated blood is pumped to the lungs, where pulmonary capillaries facilitate alveolar **gas exchange**. Here, O₂ diffuses into pulmonary capillaries while CO₂ diffuses out and is exhaled.

Most of the CO₂ entering the blood is transferred into **red blood cells** (RBCs) and combines with water to form carbonic acid (H₂CO₃), an unstable compound that rapidly dissociates to form bicarbonate (HCO₃⁻) and hydrogen (H⁺) ions. This reaction, known as the bicarbonate buffer system, is in equilibrium and helps maintain blood **pH**:



Generally, free H⁺ ions generated from this reaction bind hemoglobin and other circulating proteins, preventing blood pH from *decreasing* below the optimal level. In addition, HCO₃⁻ ions can bind H⁺ ions to reform H₂CO₃, removing additional free H⁺ ions.

The question states that the patient has decreased *pulmonary* gas exchange, indicating that O₂ uptake into blood and CO₂ removal from blood at the lungs are decreased. If left untreated, this condition will lead to *low* O₂ and **high CO₂ levels** in the patient's blood. This will shift the equilibrium of the reaction described above *to the right*, **increasing production of H₂CO₃**, which in turn will dissociate, thereby *increasing* the presence of both HCO₃⁻ and H⁺ ions in the blood. An *increased* level of circulating free H⁺ ions will lead to **decreased blood pH**, a phenomenon known as **respiratory acidosis**.

(Choice B) In this patient, decreased pulmonary gas exchange will lead to more (*not less*)
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H_2CO_3 production, which will then cause blood pH to decrease, *not* increase.

(Choices C and D) Decreased pulmonary gas exchange will cause a high blood CO_2 level, which in turn leads to increased (*not* decreased) H_2CO_3 production and a *high* level of free H^+ ions in the blood. Increased concentration of free H^+ ions will therefore decrease (*not* increase) blood pH.

Educational objective:

Pulmonary gas exchange facilitates the removal of CO_2 from blood. The amount of circulating CO_2 affects blood pH by shifting equilibrium of the bicarbonate buffer system. Impaired gas exchange in the lungs decreases the blood O_2 level and increases the blood CO_2 level, causing respiratory acidosis as H^+ concentration increases.

Time spent:0 sec

Subject:
Biology

QID:402690

System:
Circulation and Respiration

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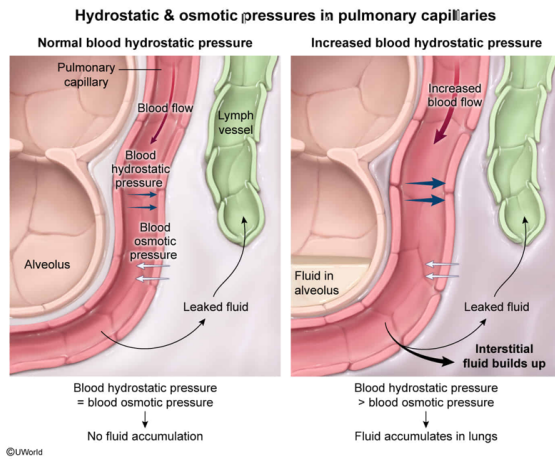
Based on the information in the passage, the excess fluid in the patient's lungs is most likely caused by which of the following at pulmonary sites of gas exchange?

- ☐ A. Increased protein concentration in the blood flowing through pulmonary capillaries [8%]
- ☐ B. Decreased solute concentration in the interstitial fluid surrounding the pulmonary capillaries [18%]
- ☒ C. Increased hydrostatic pressure within pulmonary capillaries [64%]
- ☐ D. Decreased volume of blood flowing through the pulmonary capillaries [8%]

Correct

63% answered correctly

Explanation:



According to the passage, blood flowing through vessels exerts hydrostatic pressure on the vessel walls, forcing some fluid from vessels into the interstitial space. In addition, circulating plasma proteins cause osmotic pressure (eg, solute concentration) within blood vessels to be higher than that of interstitial fluid. Fluid generally flows from areas of **low to high** osmotic pressure; therefore, blood osmotic pressure tends to draw fluid into blood vessels from the interstitial space, **opposing** hydrostatic pressure and *minimizing* blood vessel fluid loss. Additionally, **lymph vessels** generally collect any excess fluid forced from blood vessels (known as lymph when flowing through the lymphatic system), ultimately returning it to the bloodstream.

As blood flow through capillaries *increases*, hydrostatic pressure exerted on the vessel walls also *increases*, leading to a *higher* rate of fluid leakage from capillaries into interstitial space. If the rate of fluid leakage from capillaries *exceeds* the rate at which fluid is taken up by surrounding blood and lymph vessels, a higher volume of leaked fluid can build up in associated tissues (ie, edema).

The excess fluid in this patient's lungs (ie, **pulmonary edema**) is the result of an *increased* volume of blood flowing through pulmonary capillaries, which **increases hydrostatic pressure** exerted on capillary walls. High hydrostatic pressure **increases the rate of fluid leakage** into the interstitial space surrounding pulmonary capillaries, causing excess fluid accumulation in the lungs.

(Choice A) Increased protein concentration in blood flowing through pulmonary capillaries would *increase* blood osmotic pressure. Because fluid generally flows from areas of low to high osmotic pressures, higher blood osmotic pressure would tend to pull *more* fluid into pulmonary capillaries from the interstitial space, decreasing (*not* increasing) pulmonary fluid accumulation.

(Choice B) *Decreased* solute concentration in the interstitial fluid surrounding pulmonary capillaries would increase the flow of fluid *into* the capillaries (ie, region of higher solute concentration), decreasing (*not* increasing) fluid accumulation in the lungs.

(Choice D) Decreased blood volume flowing through pulmonary capillaries would cause decreased blood hydrostatic pressure, leading to decreased (*not* increased) lung fluid accumulation.

Educational objective:

Blood flowing through capillaries exerts hydrostatic pressure on the vessel walls. As a result, some fluid is forced from vessels into the interstitial space. In contrast, circulating proteins in the bloodstream cause blood osmotic pressure to be higher than that of the interstitial fluid, drawing fluid from the interstitial space into the capillaries.

Time spent:0 sec**Subject:**
Biology**QID:**402691**System:**
Circulation and Respiration